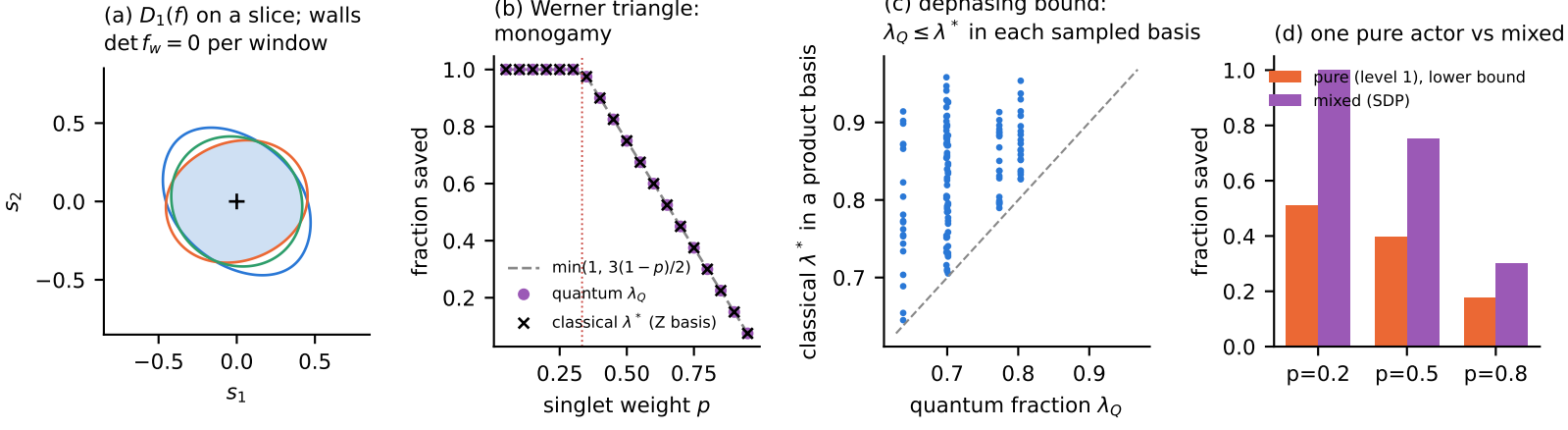
(a) $D_1(f)$ on a slice; walls
det $f_w = 0$ per window
0.5
0.0
−0.5
$s_2$
−0.5
0.0
0.5
$s_1$
(b) Werner triangle:
monogamy
1.0
0.8
0.6
0.4
0.2
fraction saved
min(1, 3(1 − p)/2)
quantum $\lambda_Q$
classical $\lambda^*$ (Z basis)
0.25
0.50
0.75
singlet weight $p$
(c) dephasing bound:
$\lambda_Q \le \lambda^*$ in each sampled basis
classical $\lambda^*$ in a product basis
0.9
0.8
0.7
0.7
0.8
0.9
quantum fraction $\lambda_Q$
(d) one pure actor vs mixed
pure (level 1), lower bound
mixed (SDP)
1.0
0.8
0.6
0.4
0.2
0.0
fraction saved
p=0.2
p=0.5
p=0.8